PROJECT REPORT

AI Smart Classroom Monitoring System Using Phone Detection

ABSTRACT

The AI Smart Classroom Monitoring System is developed to detect mobile phone usage by students during classroom hours. This system uses a webcam to capture live video and applies the YOLOv8 object detection model to identify mobile phones in real time. When a phone is detected, the system highlights it with a bounding box and triggers an alert sound to notify the teacher. The live camera feed is displayed on a web interface using Flask. This system helps teachers monitor students effectively and maintain discipline in the classroom environment.

INTRODUCTION

Artificial Intelligence (AI) and Computer Vision technologies are widely used in monitoring and security systems. In classrooms, students sometimes use mobile phones without the teacher's knowledge. This project provides an automated solution to detect mobile phone usage using AI. The system uses a webcam, Python programming, and the YOLOv8 object detection model to detect mobile phones in real time and alert the teacher through sound and visual notifications

OBJECTIVE

The main objectives of this project are:

- To detect mobile phone usage using AI.
- To monitor students using a live camera feed.
- To alert the teacher when a phone is detected.
- To display live monitoring on a website.
- To improve classroom discipline using automated monitoring.

EXISTING SYSTEM

In the existing system, classroom monitoring is done manually by teachers. Teachers cannot constantly monitor all students. Students may use phones without being noticed.

Disadvantages:

Manual monitoring is difficult.

Teacher cannot monitor continuously.

No automatic alert system.

TECHNOLOGIES USED

Programming Language

- Python

Libraries Used

- OpenCV – for camera and image processing
- YOLOv8 – for object detection
- Flask – for web application
- Winsound – for alert sound

Tools Used

- VS Code – for coding
- Webcam – for video input
- Web Browser – for display

SYSTEM ARCHITECTURE

Camera → OpenCV → YOLOv8 Model → Phone Detection
→ Alert Sound → Website Display

WORKING PROCEDURE:

- The webcam captures live video.
- The video frame is sent to YOLOv8 model.
- The model detects objects in the frame.
- If a mobile phone is detected:
- A red box is drawn.
- Alert sound is played.
- The result is displayed on the website.

FUTURE ENHANCEMENT

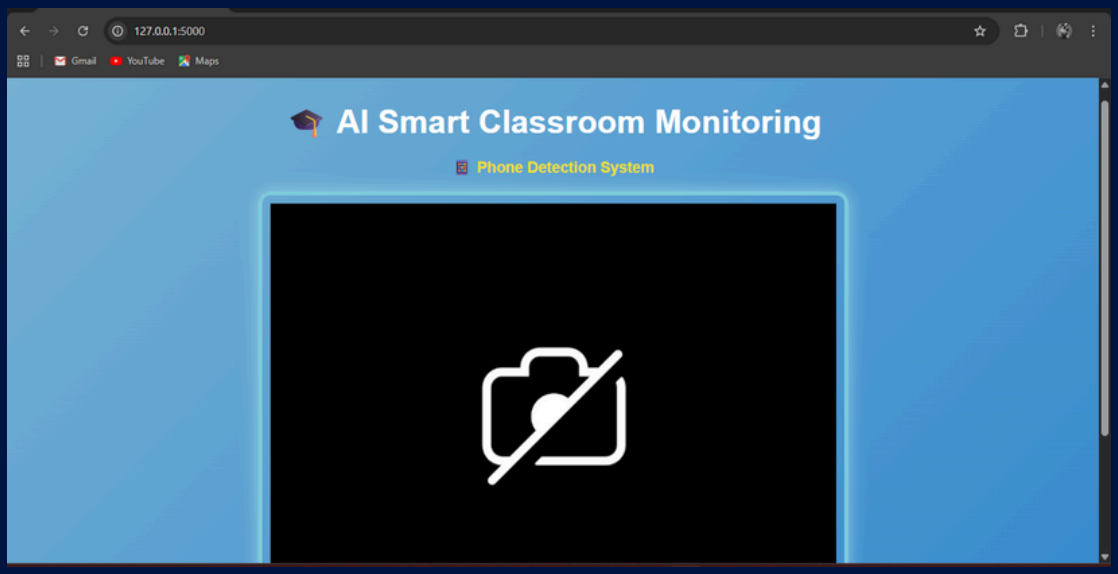
- Screenshot capture when phone detected
- Email alert system
- Multiple camera support
- Admin dashboard
- Cloud monitoring system

SOURCE CODE

app.py > ...

```python
from flask import Flask, render_template, Response
import cv2
import winsound
from ultralytics import YOLO
import threading
from playsound import playsound

app = Flask(__name__)

# load YOLO model
model = YOLO("yolov8n.pt")

# open camera
camera = cv2.VideoCapture(0)

alert_playing = False

# alert sound function
def play_alert():
    global alert_playing
    try:
        winsound.PlaySound("static/alert.wav", winsound.SND_FILENAME)
    except Exception as e:
        print("Sound error:", e)
    alert_playing = False

def generate_frames():
    global alert_playing

    while True:

        success, frame = camera.read()

        if not success:
            break

        # detect objects
        results = model(frame)
```

OUTPUT


127.0.0.1:5000
Gmail
YouTube
Maps
AI Smart Classroom Monitoring
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CONCLUSION

The AI Smart Classroom Monitoring System is an effective solution to detect mobile phone usage using AI and computer vision. It helps teachers monitor students automatically and improves classroom discipline. The system is efficient, reliable, and useful for educational institutions.